
```

1  import matplotlib.pyplot as plt
2  import numpy as np
3  import pandas as pd
4  from scipy.optimize import curve_fit
5  from pathlib import Path
6  # from chshUngl import data_alice
7
8  BASE_DIR = Path(__file__).parent
9
10 DATA_PATH = BASE_DIR / "FP66-quant" / "2026-07-20"
11 DATA_KORR_PATH = DATA_PATH / "KorrelationsMessungen"
12 DATA_RATEN_PATH = DATA_PATH / "RatenMessungen"
13 DATA_BIT_PATH = DATA_PATH / "BitMessungen"
14
15 IMAGE_PATH = BASE_DIR / "Images"
16 EINZELZ_PATH = IMAGE_PATH / 'Einzelzählraten'
17 KOINZIDENZ_PATH = IMAGE_PATH / 'Koinzidenz'
18
19
20 ### ---- Matplotlib settings ---- ###
21 plt.rcParams['axes.titlesize'] = 16
22 plt.rcParams['text.usetex'] = True
23
24 FONTSIZE = 14
25
26 ### ---- Utility Functions ---- ###
27 ROUND_TO = 4
28 def get_unsicherheit(df, key):
29     N_max = np.max(df[key])
30     N_min = np.min(df[key])
31     return (np.round(np.sqrt(N_min), ROUND_TO), np.round(np.sqrt(N_max), ROUND_TO))
32
33 def get_polaristaoinsgrad(df, key, range_start=0, range_stop=-1):
34     N_max = np.max(df[key][range_start:range_stop])
35     N_min = np.min(df[key][range_start:range_stop])
36     return np.round((N_max - N_min) / (N_max + N_min), ROUND_TO)
37
38 def get_unsicherheit_polarisationsgrad(df, key):
39     N_max = np.max(df[key])
40     N_min = np.min(df[key])
41     return np.round((2 * np.sqrt(N_max * N_min / (N_max + N_min)**3)), ROUND_TO)
42
43
44 ### ---- Curve fitting Functions ---- ###
45 def func_sin_fitt(x, a, b, c, d):
46     return a * np.sin(b * (x + c)) + d
47
48 def estimate_sin_p0(x, y, n_points=None):
49     """Estimate [a, b, c, d] starting values for a*sin(b*(x+c))+d
50 from the max/min within the first n_points of (x, y)."""
51     x = np.asarray(x)
52     y = np.asarray(y)
53     if n_points is not None:
54         x, y = x[:n_points], y[:n_points]
55
56     idx_max = np.argmax(y)
57     idx_min = np.argmin(y)
58     x_max, x_min = x[idx_max], x[idx_min]
59     y_max, y_min = y[idx_max], y[idx_min]
60
61     print(x_max, x_min, y_max, y_min)
62
63     a0 = (y_max - y_min) / 2
64     b0 = np.pi / np.abs((x_max - x_min))
65     c0 = -(x_max + x_min) / 2 # note the sign fix
66     d0 = (y_max + y_min) / 2
67
68     return [a0, b0, c0, d0]
69

```

```

70
71 err_DEGREE = 0.5
72
73 columns = [
74     'AngleAlice(deg)',
75     'AngleBob(deg)',
76     'A0',
77     'A1',
78     'B0',
79     'B1',
80     'AOB0',
81     'A1B0',
82     'AOB1',
83     'A1B1',
84 ]
85
86 data_bob = pd.read_csv(
87     filepath_or_buffer = DATA_PATH / "RatenMessungen/2026-07-20_13-56-28_Zufall_Bob.txt",
88     sep = ';',
89     comment = '#',
90     header = None,
91     names = columns
92 )
93 data_bob.sort_values('AngleBob(deg)', inplace=True, ignore_index=True)
94 data_alice = pd.read_csv(
95     filepath_or_buffer = DATA_PATH / "RatenMessungen/2026-07-20_14-06-37_Zufall_Alice.txt",
96     sep = ';',
97     comment = '#',
98     header = None,
99     names = columns
100 )
101 data_alice.sort_values('AngleAlice(deg)', inplace=True, ignore_index=True)
102 # print(data)
103
104 print(
105     f"Polaristaionsgrad Bob 0: {get_polaristaionsgrad(data_bob, 'B0')} +/-
106     ↳ {get_unsicherheit_polarisationsgrad(data_bob, 'B0')}",
107     f"nmin: {np.min(data_bob['B0'])} pm {get_unsicherheit(data_bob, 'B0')[0]}",
108     f"nmax: {np.max(data_bob['B0'])} pm {get_unsicherheit(data_bob, 'B0')[1]}"
109 )
110 print(
111     f"Polaristaionsgrad Bob 0 nur stabiler bereich: {get_polaristaionsgrad(data_bob, 'B0', 18)} +/-
112     ↳ {get_unsicherheit_polarisationsgrad(data_bob, 'B0')}",
113     f"nmin: {np.min(data_bob['B0'][: 18])} pm {get_unsicherheit(data_bob, 'B0')[0]}",
114     f"nmax: {np.max(data_bob['B0'][:18])} pm {get_unsicherheit(data_bob, 'B0')[1]}"
115 )
116 print(
117     f"Polaristaionsgrad Bob 1: {get_polaristaionsgrad(data_bob, 'B1')} +/-
118     ↳ {get_unsicherheit_polarisationsgrad(data_bob, 'B1')}",
119     f"nmin: {np.min(data_bob['B1'])} pm {get_unsicherheit(data_bob, 'B1')[0]}",
120     f"nmax: {np.max(data_bob['B1'])} pm {get_unsicherheit(data_bob, 'B1')[1]}"
121 )
122 print(
123     f"Polaristaionsgrad Alice 0: {get_polaristaionsgrad(data_alice, 'A0')} +/-
124     ↳ {get_unsicherheit_polarisationsgrad(data_alice, 'A0')}",
125     f"nmin: {np.min(data_alice['A0'])} pm {get_unsicherheit(data_alice, 'A0')[0]}",
126     f"nmax: {np.max(data_alice['A0'])} pm {get_unsicherheit(data_alice, 'A0')[1]}"
127 )
128 print(
129     f"Polaristaionsgrad Alice 1: {get_polaristaionsgrad(data_alice, 'A1')} +/-
130     ↳ {get_unsicherheit_polarisationsgrad(data_alice, 'A1')}",
131     f"nmin: {np.min(data_alice['A1'])} pm {get_unsicherheit(data_alice, 'A1')[0]}",
132     f"nmax: {np.max(data_alice['A1'])} pm {get_unsicherheit(data_alice, 'A1')[1]}"
133 )
134
135 x_values_fit = np.linspace(0, 180, 10000)
136
137 n_points=18
138 popt, pcov = curve_fit(func_sin_fitt, data_bob['AngleBob(deg)'][:n_points], data_bob['B1'][:n_points])

```

```

135
136 plt.scatter(data_bob['AngleBob(deg)'], data_bob['B1'])
137 plt.title(r'\textbf{Count rates without filtering for coincidences}')
138 plt.xlabel(r"Angle HWP  $[\sim\text{circ}]^\circ$ ", fontsize=FONTSIZE)
139 plt.ylabel(r"Counts  $\$N\$$ ", fontsize=FONTSIZE)
140 plt.savefig(
141     EINZELZ_PATH / 'EinzelzählratenEinzelBob_1.jpg',
142     dpi = 900,
143     bbox_inches="tight"
144 )
145 plt.close()
146
147
148 n_points = 18
149 popt, pcov = curve_fit(
150     func_sin_fitt,
151     data_bob['AngleBob(deg)'][0:n_points],
152     data_bob['B0'][0:n_points],
153     p0=estimate_sin_p0(
154         data_bob['AngleBob(deg)'],
155         data_bob['B0'],
156         n_points=n_points
157     ),
158     bounds=(
159         [-np.inf, 2*np.pi/150, -np.inf, -np.inf], # lower:  $a>0$ ,  $\text{period}<150^\circ$ 
160         [np.inf, 2*np.pi/60, np.inf, np.inf] # upper:  $\text{period}>60^\circ$ 
161     ),
162     method='trf',
163     max_nfev=6000
164 )
165
166 plt.scatter(data_bob['AngleBob(deg)'], data_bob['B0'])
167 plt.plot(
168     x_values_fit,
169     func_sin_fitt(x_values_fit, *popt)
170 )
171 plt.title(r'\textbf{Count rates without filtering for coincidences}')
172 plt.xlabel(r"Angle HWP  $[\sim\text{circ}]^\circ$ ", fontsize=FONTSIZE)
173 plt.ylabel(r"Counts  $\$N\$$ ", fontsize=FONTSIZE)
174 plt.savefig(
175     EINZELZ_PATH / 'EinzelzählratenEinzelBob_0.jpg',
176     dpi = 900,
177     bbox_inches="tight"
178 )
179 plt.close()
180
181 plt.scatter(data_bob['AngleBob(deg)'], data_bob['B0'])
182 plt.scatter(data_bob['AngleBob(deg)'], data_bob['B1'])
183 # plt.show()
184 plt.savefig(
185     EINZELZ_PATH / 'EinzelzählratenBeideDetektorenBob.jpg',
186     dpi = 900,
187     bbox_inches="tight"
188 )
189 plt.close()
190
191 #####
192 ## ----- Polar Coordinates ----- ##
193 plt.polar(data_bob['AngleBob(deg)'], data_bob['B1'], marker='*', linestyle='')
194 # plt.show()
195 plt.savefig(
196     EINZELZ_PATH / 'EinzelzählratenEinzelPolarBob_1.jpg',
197     dpi = 900,
198     bbox_inches="tight"
199 )
200 plt.close()
201 plt.polar(data_bob['AngleBob(deg)'], data_bob['B0'], marker='*', linestyle='')
202 # plt.show()
203 plt.savefig(
204     EINZELZ_PATH / 'EinzelzählratenEinzelPolarBob_0.jpg',

```



```

275     dpi = 900,
276     bbox_inches="tight"
277 )
278 plt.close()
279
280 #####
281 # --- Beide --- #
282 #####
283 plt.polar(data_alice['AngleAlice(deg)'], data_alice['A0'], marker='*', linestyle='')
284 plt.polar(data_alice['AngleAlice(deg)'], data_alice['A1'], marker='.', linestyle='')
285 # plt.show()
286 plt.savefig(
287     EINZELZ_PATH / 'EinzelzählratenBeideDetektoren_Polar_Alice.jpg',
288     dpi = 900,
289     bbox_inches="tight"
290 )
291 plt.close()
292
293
294
295 #####
296 # -----Korrelations messung----- #
297 #####
298 # +45
299 DATA_NAME_ALICE = '2026-07-20_15-20-46_Korrelation_Alice.txt'
300 DATA_NAME_BOB = '2026-07-20_15-11-59_Korrelation_Bob.txt'
301 data_bob_corr = pd.read_csv(
302     filepath_or_buffer = DATA_RATEN_PATH / DATA_NAME_BOB,
303     sep = ';',
304     comment = '#',
305     header = None,
306     names = columns
307 )
308 data_alice_corr = pd.read_csv(
309     filepath_or_buffer = DATA_RATEN_PATH / DATA_NAME_ALICE,
310     sep = ';',
311     comment = '#',
312     header = None,
313     names = columns
314 )
315
316
317 print('Fixiert Alice +45, Bobs detektoren:')
318 print(
319     f"Polaristaionsgrad Koinzidenz AOB0: {get_polaristaionsgrad(data_bob_corr, 'AOB0')} +/-"
320     ↪ {get_unsicherheit_polarisationsgrad(data_bob_corr, 'AOB0')}",
321     f"nmin: {np.min(data_bob_corr['AOB0'])} pm {get_unsicherheit(data_bob_corr, 'AOB0')[0]}",
322     f"nmax: {np.max(data_bob_corr['AOB0'])} pm {get_unsicherheit(data_bob_corr, 'AOB0')[1]}"
323 )
324 print(
325     f"Polaristaionsgrad Koinzidenz AOB1: {get_polaristaionsgrad(data_bob_corr, 'AOB1')} +/-"
326     ↪ {get_unsicherheit_polarisationsgrad(data_bob_corr, 'AOB1')}",
327     f"nmin: {np.min(data_bob_corr['AOB1'])} pm {get_unsicherheit(data_bob_corr, 'AOB1')[0]}",
328     f"nmax: {np.max(data_bob_corr['AOB1'])} pm {get_unsicherheit(data_bob_corr, 'AOB1')[1]}"
329 )
330 print(
331     f"Polaristaionsgrad Koinzidenz A1B0: {get_polaristaionsgrad(data_bob_corr, 'A1B0')} +/-"
332     ↪ {get_unsicherheit_polarisationsgrad(data_bob_corr, 'A1B0')}",
333     f"nmin: {np.min(data_bob_corr['A1B0'])} pm {get_unsicherheit(data_bob_corr, 'A1B0')[0]}",
334     f"nmax: {np.max(data_bob_corr['A1B0'])} pm {get_unsicherheit(data_bob_corr, 'A1B0')[1]}"
335 )
336 print(
337     f"Polaristaionsgrad Koinzidenz A1B1: {get_polaristaionsgrad(data_bob_corr, 'A1B1')} +/-"
338     ↪ {get_unsicherheit_polarisationsgrad(data_bob_corr, 'A1B1')}",
339     f"nmin: {np.min(data_bob_corr['A1B1'])} pm {get_unsicherheit(data_bob_corr, 'A1B1')[0]}",
340     f"nmax: {np.max(data_bob_corr['A1B1'])} pm {get_unsicherheit(data_bob_corr, 'A1B1')[1]}"
341 )
342 print()

```

```

341 print('Fixiert Bob +45, Alices Detektoren:')
342 print(
343     f"Polaristaionsgrad Koinzidenz AOB0: {get_polaristaoinsgrad(data_alice_corr, 'AOB0')} +/-
    ↳ {get_unsicherheit_polarisationsgrad(data_alice_corr, 'AOB0')}",
344     f"nmin: {np.min(data_alice_corr['AOB0'])} pm {get_unsicherheit(data_alice_corr, 'AOB0')[0]}",
345     f"nmax: {np.max(data_alice_corr['AOB0'])} pm {get_unsicherheit(data_alice_corr, 'AOB0')[1]}"
346 )
347 print(
348     f"Polaristaionsgrad Koinzidenz AOB1: {get_polaristaoinsgrad(data_alice_corr, 'AOB1')} +/-
    ↳ {get_unsicherheit_polarisationsgrad(data_alice_corr, 'AOB1')}",
349     f"nmin: {np.min(data_alice_corr['AOB1'])} pm {get_unsicherheit(data_alice_corr, 'AOB1')[0]}",
350     f"nmax: {np.max(data_alice_corr['AOB1'])} pm {get_unsicherheit(data_alice_corr, 'AOB1')[1]}"
351 )
352 print(
353     f"Polaristaionsgrad Koinzidenz A1B0: {get_polaristaoinsgrad(data_alice_corr, 'A1B0')} +/-
    ↳ {get_unsicherheit_polarisationsgrad(data_alice_corr, 'A1B0')}",
354     f"nmin: {np.min(data_alice_corr['A1B0'])} pm {get_unsicherheit(data_alice_corr, 'A1B0')[0]}",
355     f"nmax: {np.max(data_alice_corr['A1B0'])} pm {get_unsicherheit(data_alice_corr, 'A1B0')[1]}"
356 )
357 print(
358     f"Polaristaionsgrad Koinzidenz A1B1: {get_polaristaoinsgrad(data_alice_corr, 'A1B1')} +/-
    ↳ {get_unsicherheit_polarisationsgrad(data_alice_corr, 'A1B1')}",
359     f"nmin: {np.min(data_alice_corr['A1B1'])} pm {get_unsicherheit(data_alice_corr, 'A1B1')[0]}",
360     f"nmax: {np.max(data_alice_corr['A1B1'])} pm {get_unsicherheit(data_alice_corr, 'A1B1')[1]}"
361 )
362 print()
363 plt.scatter(data_bob_corr['AngleBob(deg)'], data_bob_corr['AOB0'])
364 plt.scatter(data_bob_corr['AngleBob(deg)'], data_bob_corr['AOB1'])
365 plt.title(r'\textbf{Coincidence count rate}' + '\n' + r' Alice HWP fixated at $+45^\circ$')
366 # plt.show()
367 plt.xlabel(r"Angle HWP $[^\circ]$", fontsize=FONTSIZE)
368 plt.ylabel(r"Counts $N$", fontsize=FONTSIZE)
369 plt.savefig(
370     KOINZIDENZ_PATH / 'Koinzidenzt hlratenAliceFixiert45.jpg',
371     dpi=900,
372     bbox_inches="tight"
373 )
374 plt.close()
375
376 plt.scatter(data_alice_corr['AngleAlice(deg)'], data_alice_corr['AOB0'])
377 plt.scatter(data_alice_corr['AngleAlice(deg)'], data_alice_corr['A1B0'])
378 plt.title(r'\textbf{Coincidence count rate}' + '\n' + r' Bob HWP fixated at $+45^\circ$')
379 plt.xlabel(r"Angle HWP $[^\circ]$", fontsize=FONTSIZE)
380 plt.ylabel(r"Counts $N$", fontsize=FONTSIZE)
381
382 # plt.show()
383 plt.savefig(
384     KOINZIDENZ_PATH / 'Koinzidenzt hlratenBobFixiert45.jpg',
385     dpi=900,
386     bbox_inches="tight"
387 )
388 plt.close()
389
390
391
392
393 # -----0/90----- #
394 DATA_NAME_ALICE = '2026-07-20_14-58-07_Korrelation_Alice.txt'
395 DATA_NAME_BOB = '2026-07-20_14-42-25_Korrelation_Bob.txt'
396 data_bob_corr = pd.read_csv(
397     filepath_or_buffer = DATA_RATEN_PATH / DATA_NAME_BOB,
398     sep = ';',
399     comment = '#',
400     header = None,
401     names = columns
402 )
403 data_alice_corr = pd.read_csv(
404     filepath_or_buffer = DATA_RATEN_PATH / DATA_NAME_ALICE,
405     sep = ';',
406     comment = '#',

```

```

407     header = None,
408     names = columns
409 )
410 print("Fixiert Alice 0/90 grad, Bobs Detektor")
411 # print('Polaristaionsgrad Koinzidenz AOB0: ', get_polaristaoinsgrad(data_bob_corr, 'AOB0'))
412 # print('Polaristaionsgrad Koinzidenz AOB1: ', get_polaristaoinsgrad(data_bob_corr, 'AOB1'))
413 # print('Polaristaionsgrad Koinzidenz A1B0: ', get_polaristaoinsgrad(data_bob_corr, 'A1B0'))
414 # print('Polaristaionsgrad Koinzidenz A1B1: ', get_polaristaoinsgrad(data_bob_corr, 'A1B1'))
415 print(
416     f"Polaristaionsgrad Koinzidenz AOB0: {get_polaristaoinsgrad(data_bob_corr, 'AOB0')} +/-"
417     ↪ {get_unsicherheit_polarisationsgrad(data_alice_corr, 'AOB0')}",
418     f"nmin: {np.min(data_bob_corr['AOB0'])} pm {get_unsicherheit(data_bob_corr, 'AOB0')[0]}",
419     f"nmax: {np.max(data_bob_corr['AOB0'])} pm {get_unsicherheit(data_bob_corr, 'AOB0')[1]}"
420 )
421 print(
422     f"Polaristaionsgrad Koinzidenz AOB1: {get_polaristaoinsgrad(data_bob_corr, 'AOB1')} +/-"
423     ↪ {get_unsicherheit_polarisationsgrad(data_alice_corr, 'AOB1')}",
424     f"nmin: {np.min(data_bob_corr['AOB1'])} pm {get_unsicherheit(data_bob_corr, 'AOB1')[0]}",
425     f"nmax: {np.max(data_bob_corr['AOB1'])} pm {get_unsicherheit(data_bob_corr, 'AOB1')[1]}"
426 )
427 print(
428     f"Polaristaionsgrad Koinzidenz A1B0: {get_polaristaoinsgrad(data_bob_corr, 'A1B0')} +/-"
429     ↪ {get_unsicherheit_polarisationsgrad(data_alice_corr, 'A1B0')}",
430     f"nmin: {np.min(data_bob_corr['A1B0'])} pm {get_unsicherheit(data_bob_corr, 'A1B0')[0]}",
431     f"nmax: {np.max(data_bob_corr['A1B0'])} pm {get_unsicherheit(data_bob_corr, 'A1B0')[1]}"
432 )
433 print(
434     f"Polaristaionsgrad Koinzidenz A1B1: {get_polaristaoinsgrad(data_bob_corr, 'A1B1')} +/-"
435     ↪ {get_unsicherheit_polarisationsgrad(data_alice_corr, 'A1B1')}",
436     f"nmin: {np.min(data_bob_corr['A1B1'])} pm {get_unsicherheit(data_bob_corr, 'A1B1')[0]}",
437     f"nmax: {np.max(data_bob_corr['A1B1'])} pm {get_unsicherheit(data_bob_corr, 'A1B1')[1]}"
438 )
439 print()
440 print()
441 print("Fixiert Bob 0/90 grad, Alices Detektor")
442 # print('Polaristaionsgrad Koinzidenz AOB0: ', get_polaristaoinsgrad(data_alice_corr, 'AOB0'))
443 # print('Polaristaionsgrad Koinzidenz AOB1: ', get_polaristaoinsgrad(data_alice_corr, 'AOB1'))
444 # print('Polaristaionsgrad Koinzidenz A1B0: ', get_polaristaoinsgrad(data_alice_corr, 'A1B0'))
445 # print('Polaristaionsgrad Koinzidenz A1B1: ', get_polaristaoinsgrad(data_alice_corr, 'A1B1'))
446 print(
447     f"Polaristaionsgrad Koinzidenz AOB0: {get_polaristaoinsgrad(data_alice_corr, 'AOB0')} +/-"
448     ↪ {get_unsicherheit_polarisationsgrad(data_alice_corr, 'AOB0')}",
449     f"nmin: {np.min(data_alice_corr['AOB0'])} pm {get_unsicherheit(data_alice_corr, 'AOB0')[0]}",
450     f"nmax: {np.max(data_alice_corr['AOB0'])} pm {get_unsicherheit(data_alice_corr, 'AOB0')[1]}"
451 )
452 print(
453     f"Polaristaionsgrad Koinzidenz AOB1: {get_polaristaoinsgrad(data_alice_corr, 'AOB1')} +/-"
454     ↪ {get_unsicherheit_polarisationsgrad(data_alice_corr, 'AOB1')}",
455     f"nmin: {np.min(data_alice_corr['AOB1'])} pm {get_unsicherheit(data_alice_corr, 'AOB1')[0]}",
456     f"nmax: {np.max(data_alice_corr['AOB1'])} pm {get_unsicherheit(data_alice_corr, 'AOB1')[1]}"
457 )
458 print(
459     f"Polaristaionsgrad Koinzidenz A1B0: {get_polaristaoinsgrad(data_alice_corr, 'A1B0')} +/-"
460     ↪ {get_unsicherheit_polarisationsgrad(data_alice_corr, 'A1B0')}",
461     f"nmin: {np.min(data_alice_corr['A1B0'])} pm {get_unsicherheit(data_alice_corr, 'A1B0')[0]}",
462     f"nmax: {np.max(data_alice_corr['A1B0'])} pm {get_unsicherheit(data_alice_corr, 'A1B0')[1]}"
463 )
464 print()
465 print()
466 plt.scatter(data_bob_corr['AngleBob(deg)'], data_bob_corr['AOB0'])
467 # plt.scatter(data_bob_corr['AngleBob(deg)'], data_bob_corr['AOB1'])
468 plt.title(r'\textbf{Coincidence count rate}' + '\n' + r' Alice HWP fixated at $0^\circ$')

```

```

469 plt.xlabel(r"Angle HWP  $0^\circ$ ", fontsize=FONTSIZE)
470 plt.ylabel(r"Counts  $N$ ", fontsize=FONTSIZE)
471
472 # plt.show()
473 plt.savefig(
474     KOINZIDENZ_PATH / 'Koinzidenzt hlratenAliceFixiert0_90_AOB0.jpg',
475     dpi=900,
476     bbox_inches="tight"
477 )
478 plt.close()
479 # plt.scatter(data_bob_corr['AngleBob(deg)'], data_bob_corr['AOB0'])
480 plt.scatter(data_bob_corr['AngleBob(deg)'], data_bob_corr['AOB1'])
481 plt.title(r'\textbf{Coincidence count rate}' + '\n' + r' Alice HWP fixated at  $0^\circ$ ')
482 plt.xlabel(r"Angle HWP  $0^\circ$ ", fontsize=FONTSIZE)
483 plt.ylabel(r"Counts  $N$ ", fontsize=FONTSIZE)
484 # plt.show()
485 plt.savefig(
486     KOINZIDENZ_PATH / 'Koinzidenzt hlratenAliceFixiert0_90_AOB1.jpg',
487     dpi=900,
488     bbox_inches="tight"
489 )
490 plt.close()
491
492 plt.scatter(data_bob_corr['AngleBob(deg)'], data_bob_corr['A1B0'])
493 plt.title(r'\textbf{Coincidence count rate}' + '\n' + r' Alice HWP fixated at  $0^\circ$ ')
494 plt.xlabel(r"Angle HWP  $0^\circ$ ", fontsize=FONTSIZE)
495 plt.ylabel(r"Counts  $N$ ", fontsize=FONTSIZE)
496 # plt.show()
497 plt.savefig(
498     KOINZIDENZ_PATH / 'Koinzidenzt hlratenAliceFixiert0_90_A1B0.jpg',
499     dpi=900,
500     bbox_inches="tight"
501 )
502 plt.close()
503
504 plt.scatter(data_bob_corr['AngleBob(deg)'], data_bob_corr['A1B1'])
505 plt.title(r'\textbf{Coincidence count rate}' + '\n' + r' Alice HWP fixated at  $0^\circ$ ')
506 plt.xlabel(r"Angle HWP  $0^\circ$ ", fontsize=FONTSIZE)
507 plt.ylabel(r"Counts  $N$ ", fontsize=FONTSIZE)
508 # plt.show()
509 plt.savefig(
510     KOINZIDENZ_PATH / 'Koinzidenzt hlratenAliceFixiert0_90_A1B1.jpg',
511     dpi=900,
512     bbox_inches="tight"
513 )
514 plt.close()
515
516 plt.scatter(data_alice_corr['AngleAlice(deg)'], data_alice_corr['AOB0'])
517 plt.scatter(data_alice_corr['AngleAlice(deg)'], data_alice_corr['A1B0'])
518 plt.title(r'\textbf{Coincidence count rate}' + '\n' + r' Bob HWP fixated at  $0^\circ$ ')
519 plt.xlabel(r"Angle HWP  $0^\circ$ ", fontsize=FONTSIZE)
520 plt.ylabel(r"Counts  $N$ ", fontsize=FONTSIZE)
521 # plt.show()
522 plt.savefig(
523     KOINZIDENZ_PATH / 'Koinzidenzt hlratenBobFixiert0.jpg',
524     dpi=900,
525     bbox_inches="tight"
526 )
527 plt.close()

```
